

POWERFACTORY

PowerFactory 2024

Technical Reference

Duplex Reactor

ElmDuplex

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

PowerFactory offers both a three-phase and a single-phase model of a duplex reactor.

The single-phase duplex reactor is a centre-tapped reactor with three terminals. It is effectively made up of two oppositely mutually coupled coils, one connected between terminal 1 and the common terminal and the second connected between terminal 2 and the common terminal.

The three-phase duplex reactor model in *PowerFactory* is assumed to be formed by three single-phase duplex reactors without any mutual coupling between different phases.

When there is a balanced power flow from the common terminal toward terminal 1 and terminal 2, the configuration with the duplex reactor offers certain advantages compared to a configuration in which terminal 1 and terminal 2 are connected to the common terminal through uncoupled reactors. The duplex reactor configuration results in lower voltage drops and lower reactive losses.

Another advantage of the duplex reactor is that during a short-circuit at e.g. terminal 1, the short-circuit contribution from terminal 2 sees an increased reactance due to the coupling.

The magnetic coupling between the two coils can be expressed in terms of the coupling coefficient. Typical values for the coupling coefficient are between 0.3 and 0.6. This means that if a current flows in the first coil, approximately 30 to 60 % of the produced flux links with the second coil.

The duplex reactor was fairly common in the past and still provides a valuable technical solution for connecting power generation to an industrial power system [1].

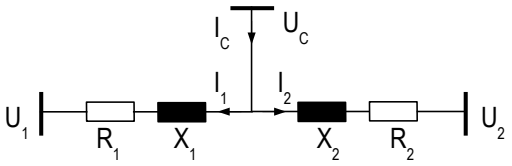


Figure 1.1: Duplex reactor configuration

2 Mathematical Modelling

The mathematical modelling of the duplex reactor has been presented in [1] and [2].

With reference to Figure 1.1, the current flowing in one coil section of the reactor will produce a voltage of opposite polarity in the second coil. The equations for the duplex reactor are described below:

$$\underline{U_1} = \underline{U_c} - \underline{Z_1} \cdot \underline{I_1} + j \cdot X_m \cdot \underline{I_2} \quad (1)$$

$$\underline{U_2} = \underline{U_c} - \underline{Z_2} \cdot \underline{I_2} + j \cdot X_m \cdot \underline{I_1} \quad (2)$$

$$\underline{I_c} = \underline{I_1} + \underline{I_2} \quad (3)$$

where

- R_1 is the resistance of coil 1
- X_1 is the self-reactance of coil 1
- $Z_1 = R_1 + j \cdot X_1$
- R_2 is the resistance of coil 2
- X_2 is the self-reactance of coil 2
- $Z_2 = R_2 + j \cdot X_2$
- $X_m = k_c \cdot \sqrt{X_1 \cdot X_2}$
- k_c is the mutual coupling coefficient

The terms $j \cdot X_m \cdot \underline{I_2}$ and $j \cdot X_m \cdot \underline{I_1}$ represent the mutual coupling between the coils.

The above equations can be rearranged differently, leading to an equivalent circuit for the duplex reactor:

$$\underline{U_c} - \underline{U_1} = R_1 \cdot \underline{I_1} + j \cdot (X_1 + X_m) \cdot \underline{I_1} + (-j \cdot X_m) \cdot (\underline{I_1} + \underline{I_2}) \quad (4)$$

$$\underline{U_c} - \underline{U_2} = R_2 \cdot \underline{I_2} + j \cdot (X_2 + X_m) \cdot \underline{I_2} + (-j \cdot X_m) \cdot (\underline{I_1} + \underline{I_2}) \quad (5)$$

The equivalent circuit representing the last equations is shown in Figure 2.1.

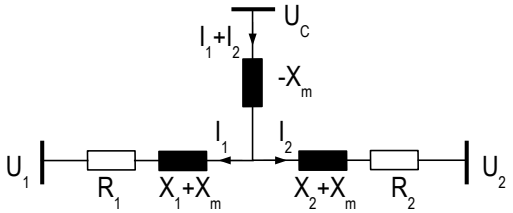


Figure 2.1: Duplex reactor equivalent circuit

2.1 Including the common path resistance

The equations presented above do not take into account the resistance R_m on the coupling path between the Common terminal and the mid-point of the reactor. This common path resistance is normally very small and it can be neglected in the vast majority of cases. The common path resistance can be taken into account according to Figure 2.2 and 2.3.

The equations of the duplex reactor developed above remain unchanged if the voltage drop over R_m is subtracted from the Common terminal voltage U_c and the internal voltage $U_{c,int}$ is used in the duplex reactor equations instead of U_c.

The duplex reactor equations can be expressed as:

$$\underline{U_{c,int}} = \underline{U_c} - R_m \cdot (\underline{I_1} + \underline{I_2}) \quad (6)$$

$$\underline{U_{c,int}} - \underline{U_1} = R_1 \cdot \underline{I_1} + j \cdot (X_1 + X_m) \cdot \underline{I_1} + (-j \cdot X_m) \cdot (\underline{I_1} + \underline{I_2}) \quad (7)$$

$$\underline{U_{c,int}} - \underline{U_2} = R_2 \cdot \underline{I_2} + j \cdot (X_2 + X_m) \cdot \underline{I_2} + (-j \cdot X_m) \cdot (\underline{I_1} + \underline{I_2}) \quad (8)$$

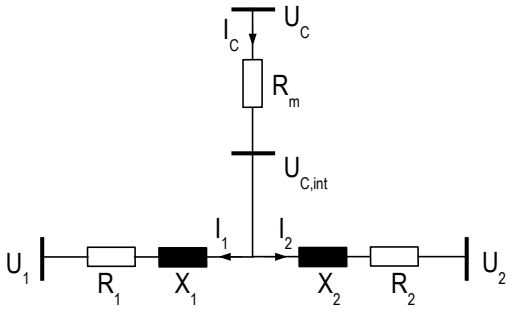


Figure 2.2: Duplex reactor configuration, including the common path resistance

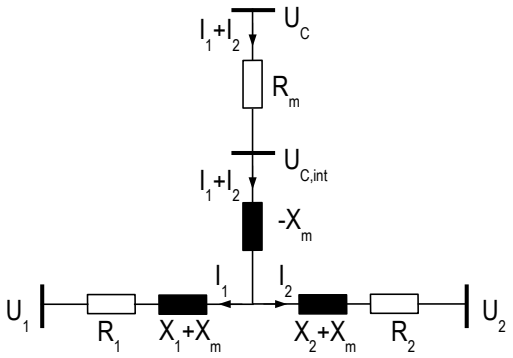


Figure 2.3: Duplex reactor equivalent circuit, including the common path resistance

3 Zero Sequence

It is possible to define a $X0toX1$ ratio in order to account for different positive and zero-sequence reactances.

Positive and zero sequence resistances are assumed to be equal.

4 Impedance Input Data

The impedance data are usually available from a measurement report.

The following impedance data are required in *PowerFactory*:

- impedance between *Common terminal* and *terminal 1*
- impedance between *Common terminal* and *terminal 2*
- mutual magnetic coupling coefficient or reactance/inductance
- common path resistance in mΩ
- zero sequence to positive sequence reactance ratio

The impedance between *Common terminal* and *terminal 1* can be obtained through a short-circuit test between the two terminals, while *terminal 2* is open. With reference to Figure 2.3, the input resistance is the sum of the coil 1 resistance (R_1) and the common path resistance (R_m) and the input reactance is the self-reactance of coil 1 (X_1):

$$rrea1 = R_1 + R_m \tag{9}$$

$$xrea1 = X_1 \tag{10}$$

The impedance between *Common terminal* and *terminal 2* can be obtained through a short-circuit test between the two terminals, while *terminal 1* is open. With reference to Figure 2.3, the input resistance is the sum of the coil 2 resistance (R_2) and the common path resistance (R_m) and the input reactance is the self-reactance of coil 2 (X_2).

$$rrea2 = R_2 + R_m \tag{11}$$

$$xrea2 = X_2 \tag{12}$$

If its value is available from the manufacturer or measurements, the common path resistance can be entered in mΩ. By default, it is neglected since it is often very small. Notice that if the common path resistance is neglected, the input resistance between *Common terminal* and *terminal 1* (or *2*) coincides with coil 1 (or 2) resistance.

5 Common Relationships

Given the coil self reactances and the mutual coupling coefficient, the mutual reactance is calculated as:

$$X_m = k_c \cdot \sqrt{X_1 \cdot X_2} \tag{13}$$

The mutual inductance is calculated as:

$$L_m = \frac{X_m}{2 \cdot \pi \cdot fnom} \tag{14}$$

If the mutual reactance is entered as data, the coupling coefficient is calculated from Equation 13 as:

$$k_c = \frac{X_m}{\sqrt{X_1 \cdot X_2}} \tag{15}$$

Between the self-inductance and the self-reactance of a coil, e.g. 1, the following relation exists:

$$L_1 = \frac{X_1}{2 \cdot \pi \cdot fnom} \tag{16}$$

The impedance data of a coil can also be entered in terms of absolute impedance (Zd) and copper losses (Pcu) or short-circuit voltage (uk) and copper losses (Pcu). The relationships between various quantities are reported below for coil 1. Similar relationships hold for coil 2.

5.1 Three-phase Model

The following relationships hold for the three-phase model:

$$Zd = \sqrt{R_1^2 + X_1^2} \tag{17}$$

$$uk = 100 \cdot \frac{\sqrt{R_1^2 + X_1^2}}{Zbase} \tag{18}$$

$$Pcu = 3 \cdot R_1 \cdot Inom^2 \tag{19}$$

where

$$Zbase = \frac{ucn^2}{Sn}$$

$$Inom = \frac{Sn}{\sqrt{3} \cdot ucn}$$

and ucn is the line-line rated voltage, Sn is the rated power.

5.2 Single-phase Model

For a single-phase duplex reactor, the relationships are as follows:

$$Zd = \sqrt{R_1^2 + X_1^2} \tag{20}$$

$$uk = 100 \cdot \frac{\sqrt{R_1^2 + X_1^2}}{Zbase} \tag{21}$$

$$Pcu = R_1 \cdot Inom^2 \tag{22}$$

where

$$Zbase = \frac{(ucn/\sqrt{3})^2}{Sn}$$

$$Inom = \frac{Sn}{ucn/\sqrt{3}}$$

6 Harmonics

It is possible to define a frequency characteristic for the following parameters: *rrea1* ($R_1 + R_m$), *lrea1* (L_1), *rrea2* ($R_2 + R_m$), *lrea2* (L_2), *lm* (L_m), *rm* (R_m).

7 References

[1] P. S. Hamer. Application of the duplex reactor — an unusual and forgotten technique to reduce short circuit duty. *IEEE Transactions on Industry Applications*, 42(2):612–617, 2006.

[2] J. Twiss J.C. Das, W.F. Robertson. Duplex reactor for a large cogeneration distribution system - an old concept reinvestigated. *TAPPI Engineering Conference Proceedings*, pages 637–648, 1991.

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